

Unit -2:

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Prepared by: *Mr. Mudit Saxena & Ms. Arpita johri*

Reviewed by: *Prof.(Dr.Kimmi Verma)*

Note: *Handouts provided are supplementary materials. Students are advised to refer the prescribed text and reference books for comprehensive understanding.*

1. Bipolar Junction Transistor (BJT)

1.1. Introduction to BJT

- A bipolar junction transistor is a three-terminal semiconductor device that consists of two p-n junctions which can amplify or magnify a signal.
- It is a current controlled device.
- The term **bipolar** reflects the current in BJT flows due to electrons as well as holes.
- The three terminals of the BJT are Base, Collector, and Emitter.

1.2. Need to Study BJT

The main basic function of a BJT is to amplify current. This allows BJTs to be used as amplifiers or switches to produce wide applicability in electronic equipment including mobile phones, industrial control, television, and radio transmitters.

1.3. Constructional Details of BJT

A transistor is basically a Si or Ge crystal containing three separate regions.

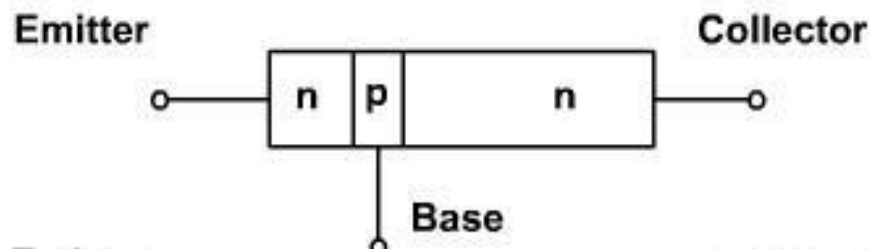


Fig. 1.3.1. Construction of BJT

- The bipolar junction transistor (BJT) is constructed with three doped semiconductor regions separated by two *p-n* junctions.
- Regions are called **Emitter**, **Base** and **Collector**.
- The depletion layers at both the junctions do not have the same width because different regions have different doping levels.

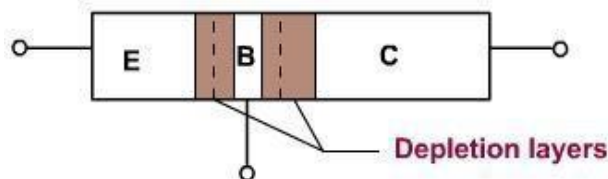
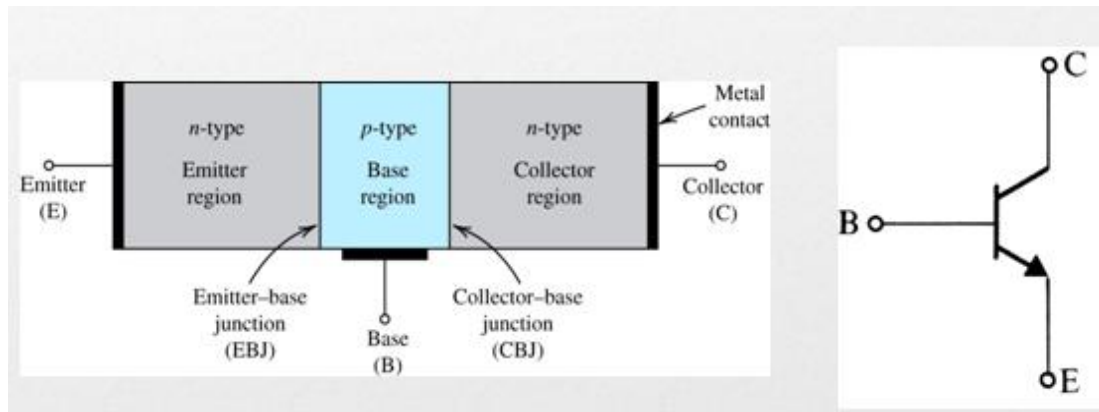


Fig. 1.3.2. Variable depletion layer in BJT

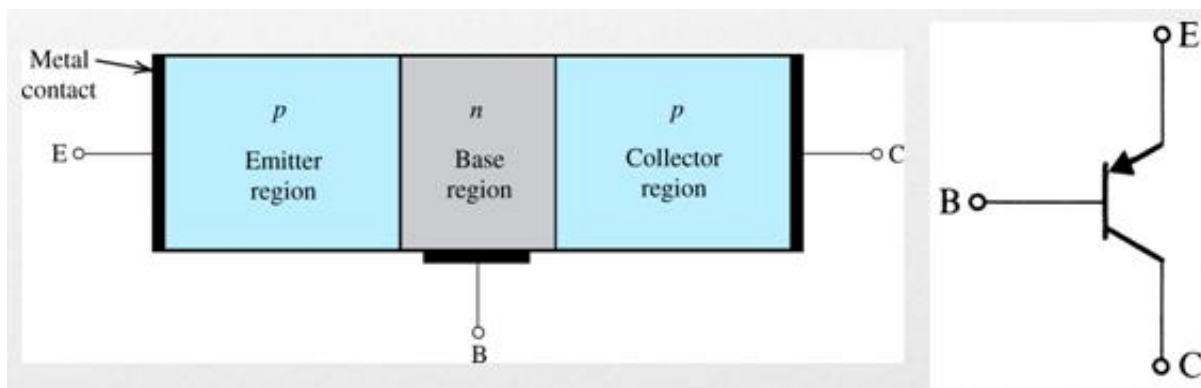
1.4. Types of BJT

BJT is of two types:

a. NPN Transistor



b. PNP Transistor



1.5. Terminals of BJT

- a. **Emitter:** Emitter acts like a source. It is heavily doped because it must provide majority charge carriers.
 - **For pnp-** The majority charge carriers are holes
 - **For npn-** The majority charge carriers are electrons
- b. **Base:** It acts as a bridge between emitter & collector. Its area is smallest among three terminals of BJT & it is lightly doped.
- c. **Collector:** It collects the majority charge carriers provided by the emitter. It is moderately doped. Its area is largest among three terminals of BJT.

1.6. Working Principle of BJT

BJT operates in one of the following three regions:

- ❖ **Cutoff region** (for digital circuits – as an OFF Switch)
[Here EB junction – Reverse Biased & CB Junction – Reverse Biased]
- ❖ **Saturation region** (for digital circuits – as an ON Switch)
[Here EB junction – Forward Biased & CB Junction – Forward Biased]
- ❖ **Active region** (for analog circuits – as an Amplifier)
[Here EB junction – Forward Biased & CB Junction – Reverse Biased]

- The basic operation in the **Active mode (i.e. BJT is configured as an amplifier)** will be described using the pnp transistor.
- Emitter Base PN junction is forward biased, whereas the Collector Base PN Junction is reverse biased.

At Emitter Base PN Junction:

- As emitter & base are forward biased, the electrons and the holes will be repelled by the biasing battery.
- This movement of electrons & holes constitute emitter current I_E (mA) and base current I_B (μ A).

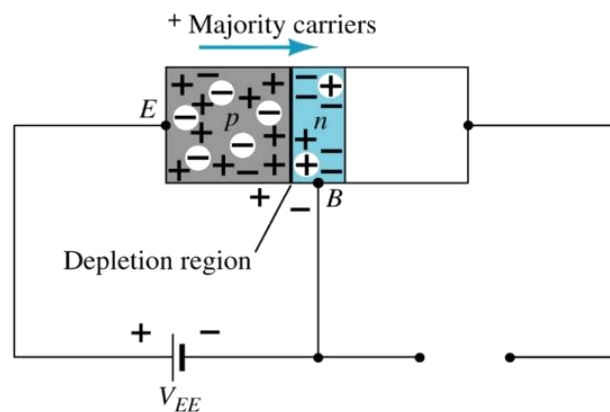


Figure 1.6.1. Movement of charge carriers at the forward biased Emitter Base junction

At Collector Base PN Junction:

- The current due to majority charge carriers is zero.
- The minority charge carriers move under the influence of the bias battery and constitute reverse saturation current.

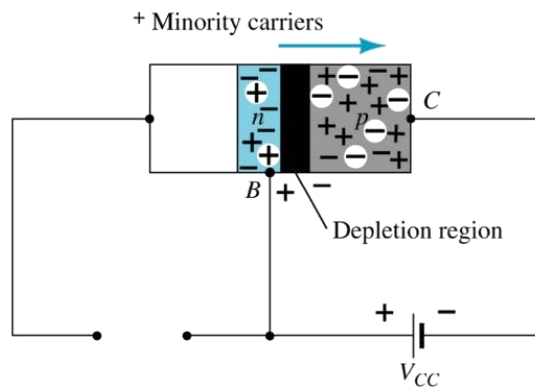


Figure 1.6.2. Movement of charge carriers at the reverse biased Collector Base junction

- The collector current comprises of two components –

$$I_C = I_{Cmajority} + I_{Cminority}$$

I_{CO} – Leakage current

$I_{Cmajority}$ – due to the charge carriers provided by the emitter via the base

- The transistor current equation is given by: $I_E = I_C + I_B$

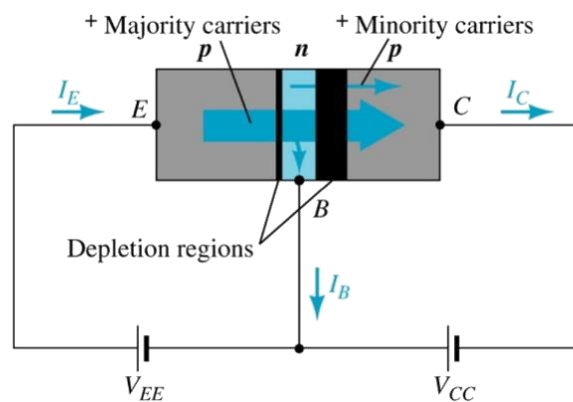


Fig. 1.6.3. Working Principle of BJT

1.7. Configurations in BJT

1.7.1. Common Base Configuration

Common-base terminology is derived from the fact that the base is common to both input and output side of the configuration.

1.7.1.1. Circuit Diagram

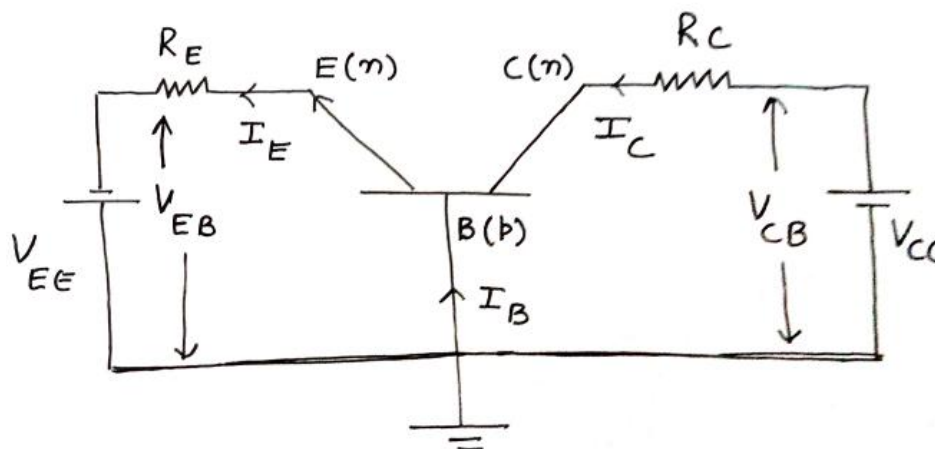


Fig. 1.7.1. Circuit Diagram of CB Configuration

1.7.1.2. Current Gain

$$\alpha = I_C (\text{mA}) / I_E (\text{mA})$$

Since input current I_E is greater than output current I_C therefore the value of α will always be less than 1, hence common base configuration is never used for amplification.

1.7.1.3. Input characteristics

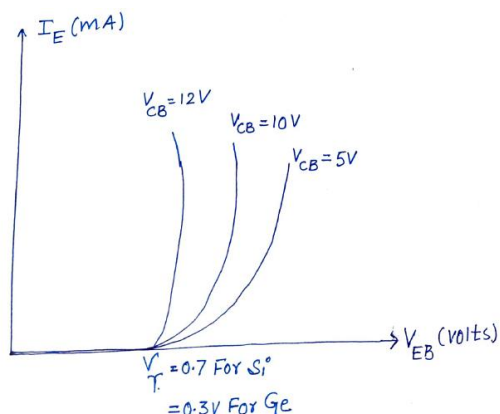


Fig. 1.7.2. Input Characteristics of CB Configuration

- Input characteristics are plotted between input current I_E & input voltage V_{EB} , keeping output voltage V_{CB} – constant.
- Input characteristics is also known as emitter curve.
- From the graph, it is clear that when input voltage $V_{EB} < V_\gamma$ (threshold voltage), input current $I_E = 0$ but when $V_{EB} > V_\gamma$, input current I_E flows exponentially.

1.7.1.4. Output characteristics

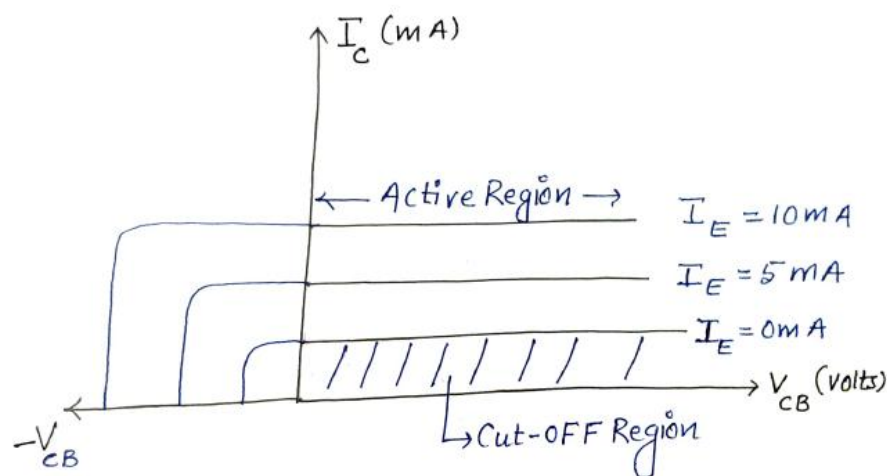


Fig. 1.7.3. Output Characteristics of CB Configuration

- Output characteristics are plotted between output current I_C & output voltage V_{CB} , keeping input current I_E – constant.
- Output characteristics is also known as collector curve.
- The output characteristics have 3 basic regions:
 - Active region** – The flat curve shows that no amplification is taking place.
 - Cutoff region** – If input current I_E is zero then output current I_C must be zero but from the characteristics, we can see that I_E is zero, but I_C is not equal to zero, a small amount of output current (I_C) flows. This output current is due to minority charge carriers (as both the junctions are reverse biased), called leakage current & this region is known as cut-off region.
 - Saturation region** – In this region, both the junctions become forward biased, and the output current doesn't depend on input current.

1.7.2. Common Emitter Configuration

Common-Emitter terminology is derived from the fact that Emitter is common or reference to both input and output sides of the configuration.

1.7.2.1. Circuit Diagram

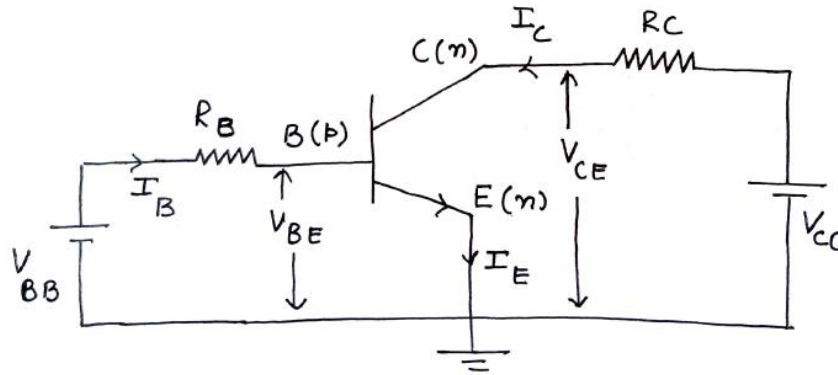


Fig. 1.7.4. Circuit Diagram of CE Configuration

1.7.2.2. Current Gain

$$\beta = I_C (\text{mA}) / I_B (\mu\text{A})$$

Since, β is very large in value therefore output current I_C is much larger than input current I_B , this is one of the major reasons that why CE configuration is always preferred for amplification.

1.7.2.3. Input characteristics

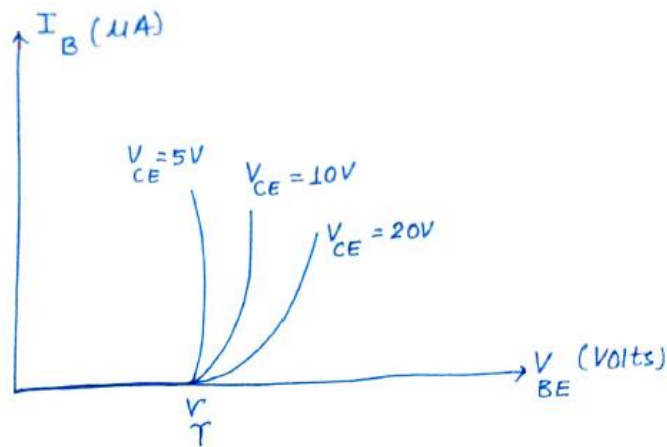


Fig. 1.7.5. Input Characteristics of CE Configuration

- Input characteristics are plotted between input current I_B & input voltage V_{BE} , keeping output voltage V_{CE} – constant.
- Input characteristics are also known as base curves.
- From the graph, it is clear that when input voltage $V_{BE} < V_\gamma$ (threshold voltage), input current $I_B = 0$ but when $V_{BE} > V_\gamma$, input current I_B flows exponentially.
- Input resistance of CE configuration is of the order of megaohms, due to very high input impedance, CE configuration is always preferred for amplification.

1.7.2.4. Output characteristics

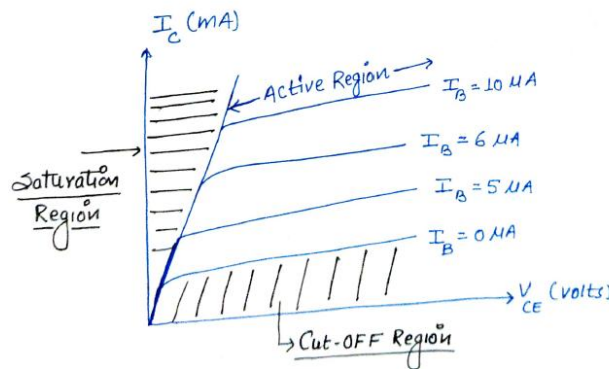


Fig. 1.7.6. Output Characteristics of CE Configuration

- Output characteristics are plotted between output current I_C & output voltage V_{CE} , keeping input current I_B – constant.
- Output characteristics are also known as collector curves.
- The output characteristics have 3 basic regions:
 - Active region** – The upward going curve shows that faithful amplification is taking place.
 - Cutoff region** – If input current I_B is zero then output current I_C must be zero but from the characteristics, we can see that I_B is zero, but I_C is not equal to zero, a small amount of output current (I_C) flows. This output current is due to minority charge carriers (as both the junctions are reverse biased), called leakage current & this region is known as cut-off region.
 - Saturation region** – In this region, both the junctions become forward biased, and the output current doesn't depend on input current.

1.7.3. Common Collector Configuration

Common-Collector terminology is derived from the fact that collector is common or reference to both the input and output sides of the configuration.

1.7.3.1. Circuit Diagram

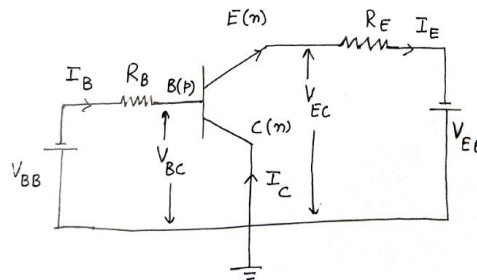


Fig. 1.7.7. Circuit Diagram of CE Configuration

1.7.3.2. Current Gain

$$\gamma = I_E (\text{mA}) / I_B (\mu\text{A})$$

1.7.3.3. Input characteristics

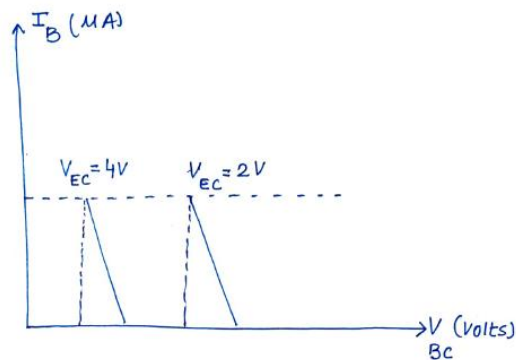


Fig. 1.7.8. Input Characteristics of CC Configuration

- Input characteristics are plotted between input current I_B & input voltage V_{BC} , keeping output voltage V_{EC} – constant.
- Input characteristics are also known as base curves.
- Nearly straight lines of the curve shows that the input resistance of CC configuration is very high; of the order of mega-ohms.

1.7.3.4. Output characteristics

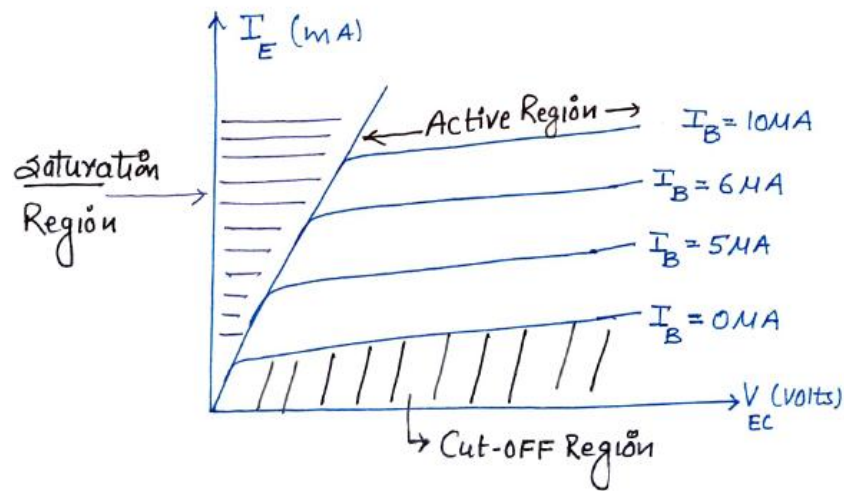


Fig. 1.7.8. Output Characteristics of CC Configuration

- Output characteristics are plotted between output current I_E & output voltage V_{EC} , keeping input current I_B – constant.
- Output characteristics are also known as emitter curves.
- The output characteristics have 3 basic regions:
 - a. Active region** –The upward going curves show that faithful amplification is taking place.
 - b. Cutoff region** – If input current I_B is zero then output current I_E must be zero but from the characteristics, we can see that I_B is zero, but I_E is not equal to zero, a small amount of output current (I_E) flows. This output current is due to minority charge carriers (as both the junctions are reverse biased), called leakage current & this region is known as cut-off region.
 - c. Saturation region**- In this region, both the junctions become forward biased and the output current doesn't depend on input current.

1.7.4. Relation between the current gains of CB & CE Configurations

We know $I_C = \alpha I_E$ — ①

but $I_E = I_B + I_C$, put in Eqⁿ ①

$$\Rightarrow I_C = \alpha (I_B + I_C)$$

$$\Rightarrow I_C = \alpha I_B + \alpha I_C$$

$$\Rightarrow I_C (1 - \alpha) = \alpha I_B$$

$$\Rightarrow \frac{I_C}{I_B} = \frac{\alpha}{1 - \alpha}$$

$$\boxed{\beta = \frac{\alpha}{1 - \alpha}} \longrightarrow \text{②}$$

From Equation 2 :- $\beta (1 - \alpha) = \alpha$

$$\Rightarrow \beta - \alpha \beta = \alpha$$

$$\Rightarrow \alpha (1 + \beta) = \beta$$

$$\boxed{\alpha = \frac{\beta}{1 + \beta}}$$

1.7.5. Relation between the current gains of CB, CE and CC Configurations

We know $I_E = I_B + I_C \longrightarrow \textcircled{1}$

Divide Eqⁿ ① by I_B :-

$$\frac{I_E}{I_B} = \frac{I_B}{I_B} + \frac{I_C}{I_B}$$

$$\boxed{\gamma = 1 + \beta} \quad \left\{ \gamma = \frac{I_E}{I_B} \text{ \& } \beta = \frac{I_C}{I_B} \right\}$$

Now divide Eqⁿ ① by I_E :-

$$\frac{I_E}{I_E} = \frac{I_B}{I_E} + \frac{I_C}{I_E}$$

$$1 = \frac{1}{\gamma} + \alpha \quad \left\{ \alpha = \frac{I_C}{I_E} \right\}$$

$$\boxed{\gamma = \frac{1}{1 - \alpha}}$$

2. Field Effect Transistors (FET)

2.1. Introduction to Field Effect Transistors

Field-Effect Transistor (FET) is a semiconductor device that consists of a channel made of a semiconductor material, with two electrodes connected at either end, namely the drain and the source. The flow of current between the source and the drain terminals is controlled by a third electrode, known as the gate, which is placed in close proximity to the channel.

- FETs have three terminals:
 - The first terminal is the source (S), through which current enters the device, denoted by I_S .
 - The second terminal is the drain (D), through which current leaves the device, denoted by I_D . The voltage between drain and source is V_{DS} .
 - The third terminal is the gate (G), which modulates the channel conductivity. By applying a voltage at the gate, I_D can be controlled
- Key Characteristics
 - - Voltage-Controlled Device: Requires minimal input current at the gate.
 - - High Input Impedance: Makes it suitable for high-impedance signal sources.
 - - Unipolar Operation: Uses either electrons or holes as carriers, resulting in lower noise.

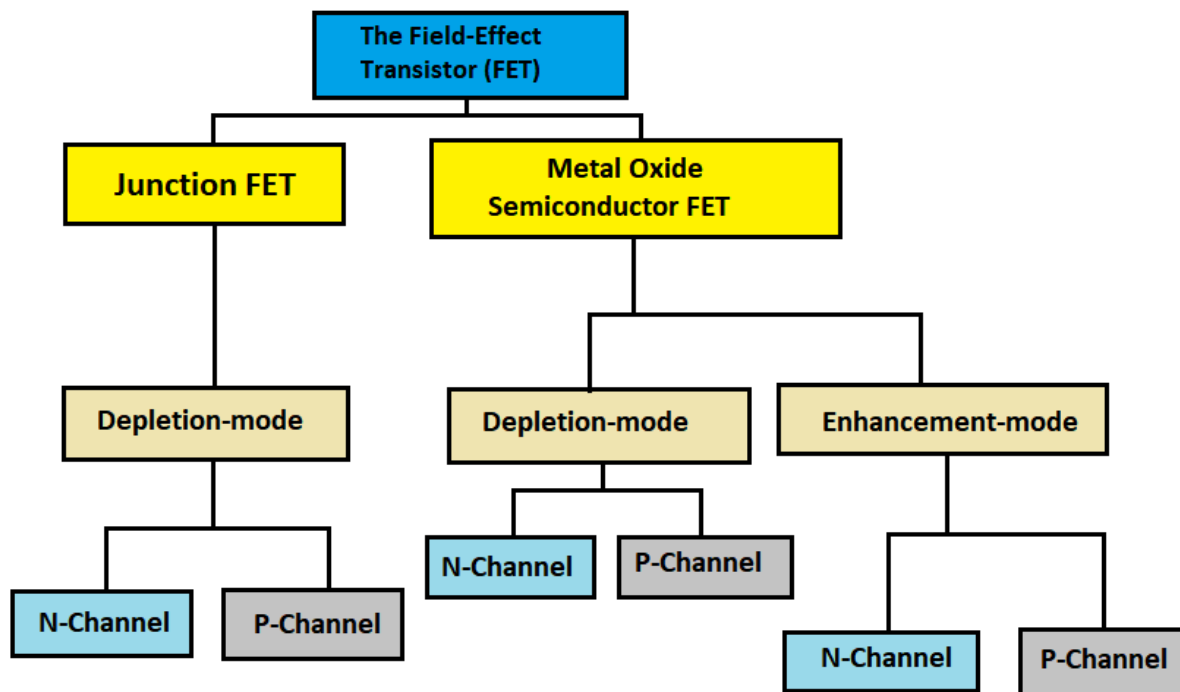


Fig. 2.1. Types of FET

2.2. Differences between BJT and FET

Feature	FET (Field Effect Transistor)	BJT (Bipolar Junction Transistor)
Type of Device	Voltage-controlled device	Current-controlled device
Operation Principle	Uses an electric field to control current	Uses input current to control the output current
Input Impedance	High (due to gate insulation)	Low
Output Impedance	Low	High
Current Flow	Majority carriers (unipolar)	Both majority and minority carriers (bipolar)
Speed	Generally faster due to high input impedance	Slower compared to FETs
Power Consumption	Low (requires less input current)	High (requires significant base current)
Thermal Stability	High (less prone to thermal runaway)	Low (prone to thermal runaway without proper compensation)
Noise Level	Low (better for high-frequency applications)	Higher (more noise compared to FETs)
Applications	Used in analog switches, amplifiers, and digital circuits	Used in power amplifiers, oscillators, and analog circuits

2.3. Advantages of FET over BJT

- High input impedance (Linear AC amplifier system)
- Temperature is more stable in FET than BJT
- Smaller than BJT
- Can be fabricated with fewer processing
- Usually use as logic switch

2.4. Types of Junction Field Effect Transistor (JFET)

A JFET consists of a thin bar of semiconductor material, which can be silicon or gallium arsenide. Regardless of whether it is an N-channel or P-channel JFET, the construction principles are similar:

- **Semiconductor Bar:** A bar of semiconductor material (N-type for N-channel and P-type for P-channel) forms the core structure.
- **P-N Junctions:** Two small regions of the opposite type (P-type for N-channel and N-type for P-channel) are diffused into the bar to create PN junctions.
- **Gate Terminal:** The two diffused regions are internally connected to form the gate terminal, creating a single control point for the device.
- **Ohmic Contacts:** The ends of the semiconductor bar form ohmic contacts with the source and drain terminals, ensuring efficient current flow.



Fig. 2.4.1. Schematic Representation of FET

2.4.1. N-Channel JFET

- Major structure is n-type material (channel) between embedded p-type material to form 2 p-n junctions.
- In the normal operation of an n-channel device, the Drain (D) is positive with respect to the Source (S). Current flows into the Drain (D), through the channel, and out of the Source (S).
- Electrons injected to Source (S) through p-type channel and flowed to Drain (D). I_D increases for positive V_{GS} .

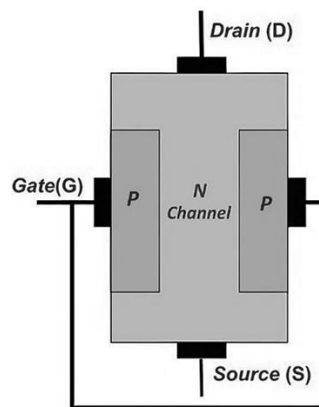


Fig. 2.4.2. N-Channel JFET

2.4.2. P-Channel JFET

- Major structure is p-type material (channel) between embedded n-type material to form 2 p-n junctions.
- Current flow: from Source (S) to Drain (D)
- Holes injected to Source (S) through p-type channel and flowed to Drain (D). I_D increases for negative V_{GS} .

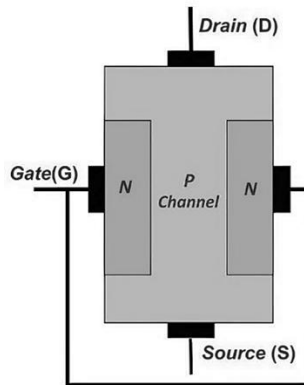


Fig. 2.4.2. P-Channel JFET

2.5. Operation of N-channel JFET

CASE1: No Gate Voltage ($V_{GS} = 0$), V_{DS} (POSITIVE)

- When a positive voltage (V_{DS}) is applied, current flows from the drain to the source through the n-channel. Drain is RB & Source is FB therefore there will be force of attraction at drain & repulsion at source causing electrons to move from source to drain generating current i.e., $I_D = I_S$

CASE 2: ($V_{GS} = 0$), V_{DS} (INCREASED POSITIVELY)

- A positive voltage applied at the drain increases the force of attraction increasing free electrons. This causes a **depletion region** to form around the gate. The depletion region reduces the effective width of the n-channel, restricting current flow. The channel is narrow as depletion layer increases, and maximum current (I_{DSS}) flows.

CASE 3: Applying Negative Gate Voltage ($V_{GS} < 0$)

- The higher the negative V_{GS} , the more the channel is pinched. As V_{GS} becomes sufficiently negative, the depletion regions on both sides of the gate expand and completely close the channel. At this point, the current flow (I_D) through the channel stops. The V_{GS} at which this happens is called the **pinch-off voltage ($-V_P$)** & **this condition is called Pinch-Off Condition**. I_D will decrease.

CASE 4: Applying Negative Gate Voltage ($V_{GS} > 0$)

- When V_{GS} is increased further (beyond V_P), the current I_D starts to increase further by pinching through the depletion layer, widening the channel.

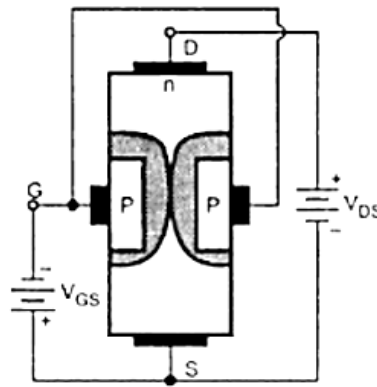


Fig. 2.5.1. Channel for pinch-off condition.

2.6. Transfer Characteristics

- In JFET, the relationship between V_{GS} (input voltage) and I_D (output current) is used to define the transfer characteristics. It is called as Shockley's Equation:

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2 \quad V_P = V_{GS(\text{OFF})}$$

- The relationship is more complicated (and not linear). As a result, FETs are often referred to a square law device.
- Obtaining transfer characteristic curve axis point from Shockley:

When $V_{GS} = 0 \text{ V}$, $I_D = I_{DSS}$

When $V_{GS} = V_{GS(\text{off})}$ or V_P , $I_D = 0 \text{ mA}$

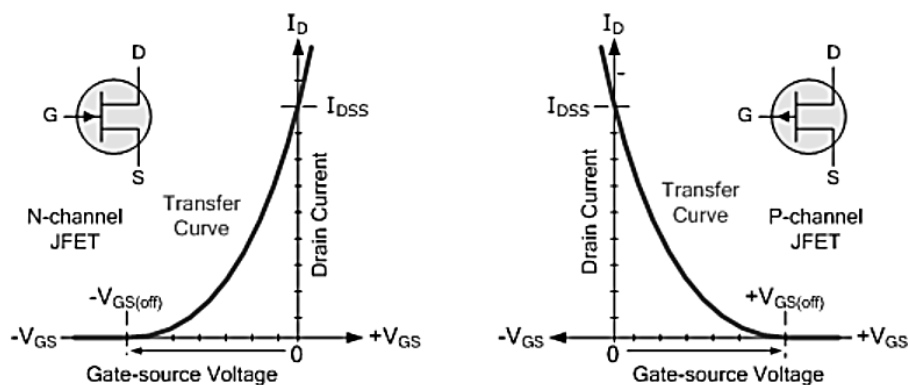


Fig. 2.6.1. Transfer characteristics of N & P type JFET

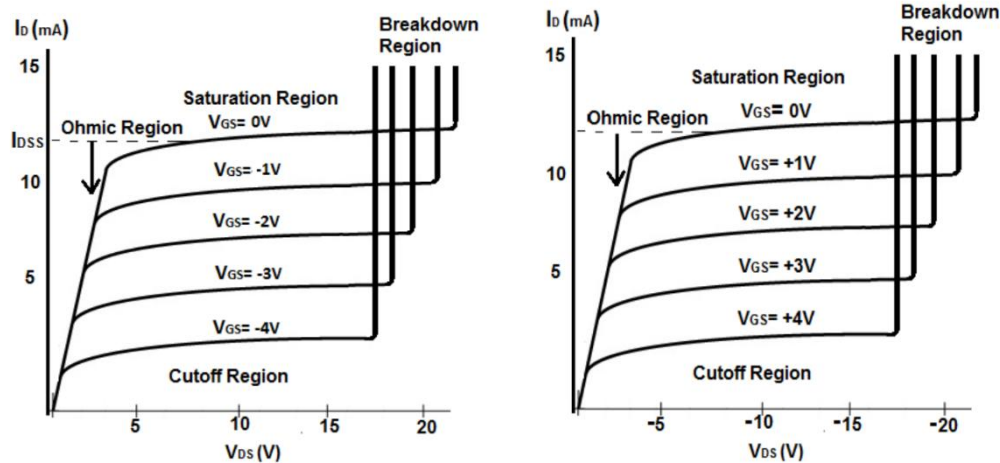


Fig. 2.6.2. Drain characteristics of N & P type JFET

2.7. JFET Characteristic curve

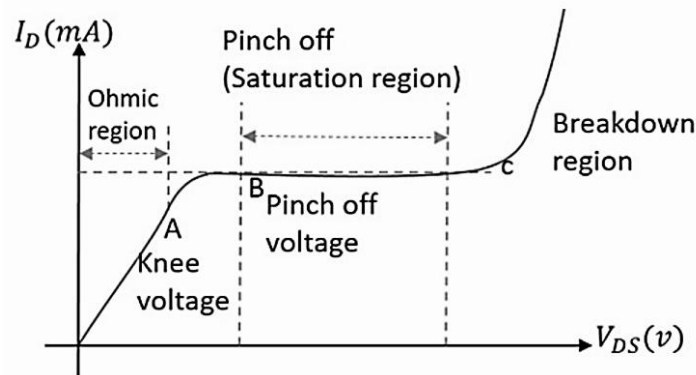


Fig. 2.7. JFET Characteristic Curve

The JFET has different characteristics in different regions, depending on the input voltages:

- **Ohmic region**

The first part of the characteristic curve where the drain current increases rapidly. In this region, the JFET acts as a **voltage-controlled resistor or voltage variable resistor (VVR)**

- **Saturation region**

Also known as the active region. In this region **JFET acts as a constant-current source**. In this region, the channel acts as a good conductor.

- **Cutoff region**

Also known as the pinched-off region, the JFET is in OFF condition and no drain current flows.

- **Breakdown region**

the transistor fails to resist the current and the current flows from drain terminal to source terminal. If this current is not limited, the JFET may be damaged.

- JFET is biased with two voltage sources: V_{DD} and V_{GG}
- V_{DD} generates voltage bias between Drain (D) and Source (S) – V_{DS} .
- V_{DD} causes drain current, I_D flows from Drain (D) to Source (S).
- V_{GG} generate voltage bias between Gate (G) and Source (S) with negative polarity source is connected to the Gate Junction (G) – reverse-biases the gate; therefore, gate current, $I_G = 0$.
- V_{GG} is to produce depletion region in N channel so that it can control the amount of drain current, I_D that flows through the channel.

2.8. MOSFET

- MOSFET is a voltage-controlled field effect transistor that differs from a JFET in that it has a “Metal Oxide” Gate electrode which is electrically insulated from the main semiconductor n-channel or p-channel by a very thin layer of insulating material usually silicon dioxide.
- The isolation of the controlling Gate makes the input resistance of the MOSFET extremely high in the Mega-ohms ($M\Omega$) region thereby making it almost infinite.
- As the Gate terminal is isolated from the main current carrying channel “NO current flows into the gate” and just like the JFET, the MOSFET also acts like a voltage-controlled resistor where the current flowing through the main channel between the Drain and Source is proportional to the input voltage.
- MOSFETs are three terminal devices with a Gate, Drain and Source and both P-channel (PMOS) and N-channel (NMOS) MOSFETs are available. In n-Channel MOS: Majority carriers are electrons & in p-Channel MOS: Majority carriers are holes.
- The MOSFET’s are available in two basic forms:
 1. **Depletion Type** – The transistor requires the Gate-Source voltage, (V_{GS}) to switch the device “OFF”. The depletion mode MOSFET is equivalent to a “Normally Closed” switch. Channel is present.
 2. **Enhancement Type** – the transistor requires a Gate-Source voltage, (V_{GS}) to switch the device “ON”. The enhancement mode MOSFET is equivalent to a “Normally Open” switch. Absence of channel.

2.8.1 Basic MOSFET structure and symbol

- The MOSFET have an additional terminal called the Substrate and it is not normally used as either an input or an output connection but instead it is used for grounding the substrate.
- The gate electrode is placed on top of a very thin insulating layer and there are a pair of small n-type regions just under the drain and source electrodes.
- If this is a solid unbroken line then this represents a “Depletion” (normally-ON) type MOSFET as drain current can flow with zero gate potential.

- If the channel line is shown dotted or broken it is an “Enhancement” (normally-OFF) type MOSFET as zero drain current flows with zero gate potential. The direction of the arrow indicates whether the conductive channel is a p-type or an n-type semiconductor device.

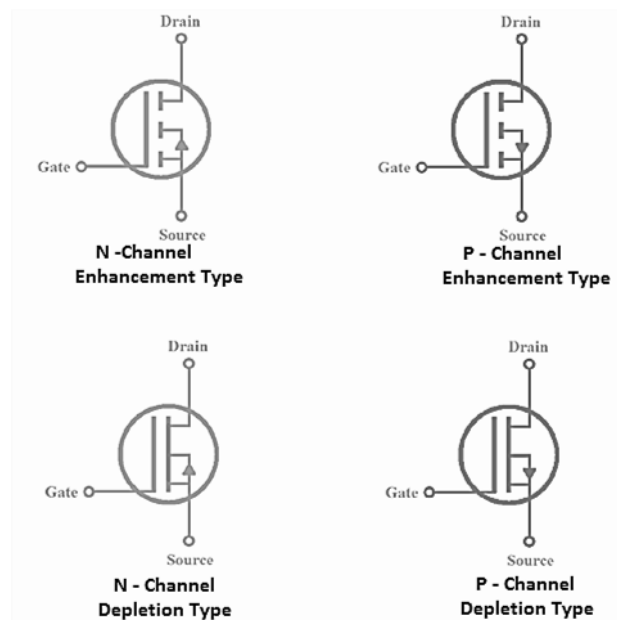


Fig 2.81. Symbols of MOS transistors.

2.8.2 Working of MOSFET

The operation of a MOSFET is based on voltage applied to the **gate terminal**, which controls the flow of current between the **source** and **drain**.

1. No Gate Voltage ($V_{GS} = 0$):

- For an n-channel MOSFET, no conductive channel exists between the source and drain.
- The device is off, and no current flows.

2. Positive Gate Voltage (n-channel MOSFET):

- A positive voltage at the gate attracts electrons towards the gate oxide layer, forming a conductive channel (inversion layer) between the source and drain.
- The channel allows current to flow from drain to source when a voltage is applied between them (V_{DS}).

3. Negative Gate Voltage (p-channel MOSFET):

- A negative voltage at the gate attracts holes, forming a conductive channel.
- Current flows from source to drain under an applied V_{DS} .

4. Gate Threshold Voltage (V_{TH}):

- A minimum gate-to-source voltage (V_{GS}) is required to create the conductive channel. This is known as the threshold voltage.

Modes of Operation

1. Cut-off Region:

- $V_{GS} < V_{TH}$ (for n-channel).
- The MOSFET is off, and no current flows.

2. Linear (Ohmic) Region:

- $V_{GS} > V_{TH}$, and V_{DS} is small.
- The MOSFET behaves like a resistor, and current increases linearly with V_{DS} .

3. Saturation (Active) Region:

- $V_{GS} > V_{TH}$, and V_{DS} is large enough.
- The MOSFET behaves like a current source, and current is determined by V_{GS} .

2.8.3 Depletion mode MOSFET

- In D-MOSFET, since the channel is present, channel conducts when $V_{GS} = 0$ making it a “normally closed” device. The circuit symbol shown above for a depletion MOS transistor uses a solid channel line to signify a normally closed conductive channel.
- For the n-channel depletion MOS transistor, a negative gate-source voltage, $-V_{GS}$ will deplete (hence its name) the conductive channel of its free electrons switching the transistor “OFF.” Likewise for p-channel depletion MOS transistor a positive gate source voltage, $+V_{GS}$ will deplete the channel of its free holes turning it “OFF.”
- In other words, for an n-channel depletion mode MOSFET: $+V_{GS}$ means more electrons and more current. While a $-V_{GS}$ means less electrons and less current. The opposite is also true for the p-channel types. Then the depletion mode MOSFET is equivalent to a “normally closed” switch.

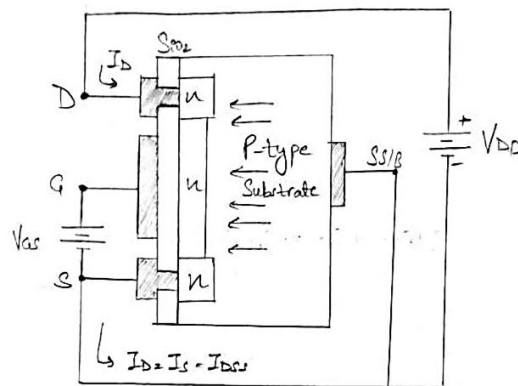


Fig.10: Basic MOSFET structure (Depletion-MOS)

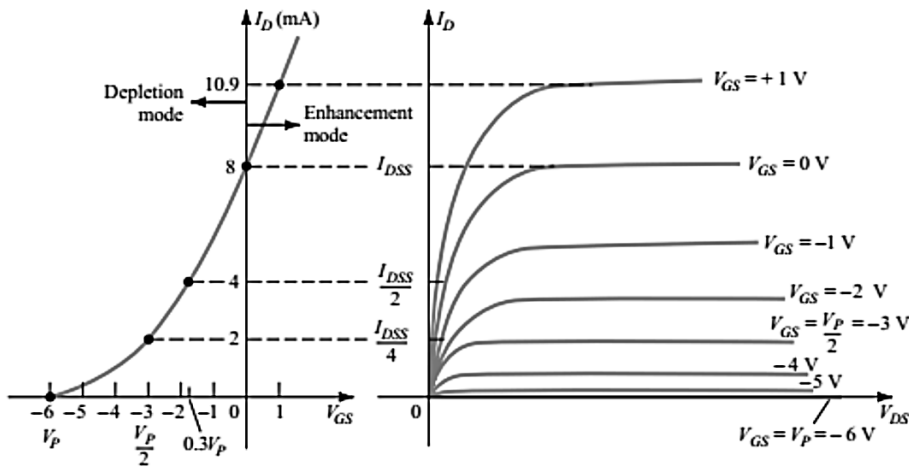


Fig 11: Transfer & Drain characteristics for N type Depletion MOSFET

Likewise, for p-channel depletion MOS transistor a positive gate source voltage, $+V_{GS}$ will deplete the channel of its free holes turning it "OFF." Also, the polarity of Drain to source is reversed, i.e., negative terminal of battery is connected to Drain & positive terminal is connected to Source.

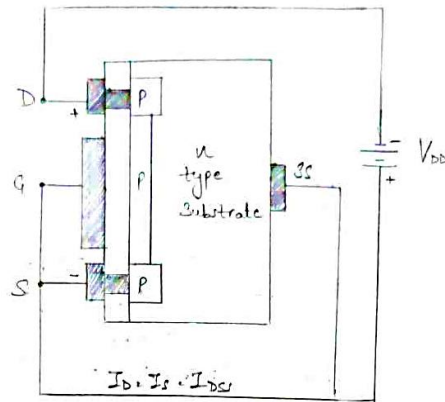


Fig.12: Basic MOSFET structure (P TYPE Depletion-MOS)

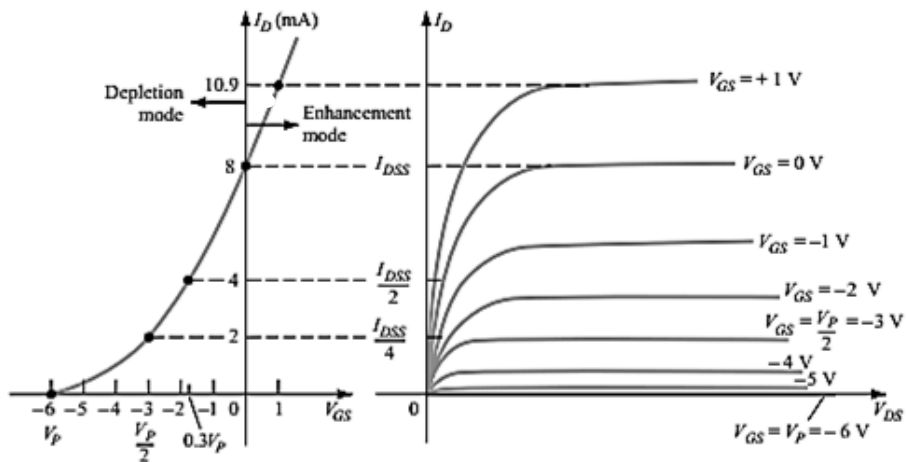


Fig 13: Transfer & Drain characteristics for P type Depletion MOSFET

2.8.3 Enhancement Mode MOSFET

- In E-MOSFET, since the channel is absent but induced due to gate source voltage.
- This results in the device being normally “OFF” (non-conducting) when the gate bias voltage, V_{GS} is equal to zero.
- For the n-channel enhancement MOS transistor a drain current will only flow when a gate voltage (V_{GS}) is applied to the gate terminal greater than the threshold voltage (V_T) level in which conductance takes place making it a transconductance device.

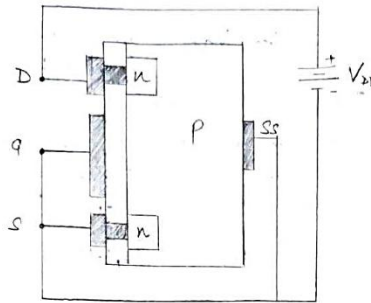


Fig.14: Basic MOSFET structure (N TYPE Depletion-MOS)

- The application of a positive (+ve) gate voltage to n-type E-MOSFET attracts more electrons towards the oxide layer around the gate thereby increasing or enhancing (hence its name) the thickness of the channel allowing more current to flow.
 - This is why this kind of transistor is called an enhancement mode device as the application of a gate voltage enhances the channel. Increasing this positive gate voltage will cause the channel resistance to decrease further causing an increase in the drain current, I_D through the channel. In other words, for an n-channel enhancement mode MOSFET: $+V_{GS}$ turns the transistor “ON,” while a zero or $-V_{GS}$ turns the transistor “OFF”.
 - Then, the enhancement-mode MOSFET is equivalent to a “normally-open” switch. The reverse is true for the p-channel enhancement MOS transistor. When $V_{GS} = 0$ the device is “OFF” and the channel is open. The application of a negative (-ve) gate voltage to the p-type E-MOSFET enhances the channels conductivity turning it “ON.” Then for an p-channel enhancement mode MOSFET: $+V_{GS}$ turns the transistor “OFF,” while $-V_{GS}$ turns the transistor “ON.”
- (a) At $V_{DS} = 0$, thermal equilibrium exists in the inverted channel region and drain current $I_D = 0$. Now if small drain voltage, $V_{DS} > 0$ is applied, a drain current proportional to the V_{DS} will start to flow from source to drain through the channel.
- (b) The channel gives a continuous path for the flow of current from source to drain. This mode of operation is called **linear/ohmic region**. The cross-sectional view of an n-channel MOSFET, operating in linear region, is shown in the figure given below.

- (c) Now if the V_{DS} is increased, charges in the channel and channel depth decrease at the end of drain. For $V_{DS} = V_{DSAT}$, the charges in the channel reduces to zero, which is called **pinch – off point**. For $V_{DS} > V_{DSAT}$, a depleted surface forms near to drain, and by increasing the drain voltage this depleted region extends to source.

This mode of operation is called **Saturation region**. The electrons coming from the source to the channel end, enter in the drain – depletion region and are accelerated towards the drain in high electric field.

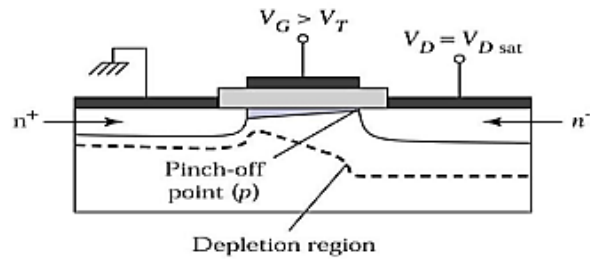


Figure 15: Pinch off Condition

P-type enhancement MOSFET

- The transfer characteristics of p-type enhancement MOSFETs from which it is evident that I_D remains zero (cutoff state) until V_{GS} becomes equal to $-V_T$. This is because, only then the channel will be formed to connect the drain terminal of the device with its source terminal.

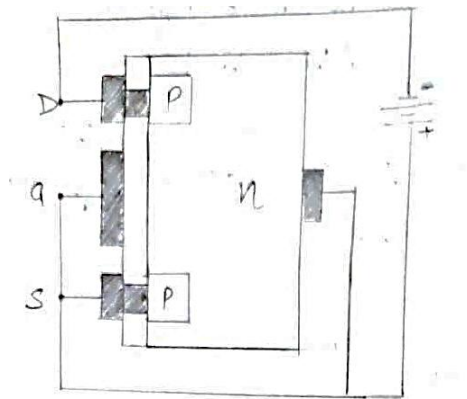


Fig.16: Basic MOSFET structure (P TYPE Depletion-MOS)

- After this, the I_{DSS} is seen to increase in reverse direction (meaning an increase in I_D , signifying an increase in the device current which will flow from source to drain) with the decrease in the value of V_{DS} .
- However, as V_{DS} becomes equal to $-V_P$, the device enters into saturation during which a saturated amount of current (I_{DSS}) flows through the device, as decided by the value of V_{GS} .

- Further it is to be noted that the value of saturation current flowing through the device is seen to increase as the V_{GS} becomes more and more negative. In addition, from the locus of the pinch-off voltage it is also clear that as V_{GS} becomes more and more negative, even the negativity of V_P also increases.

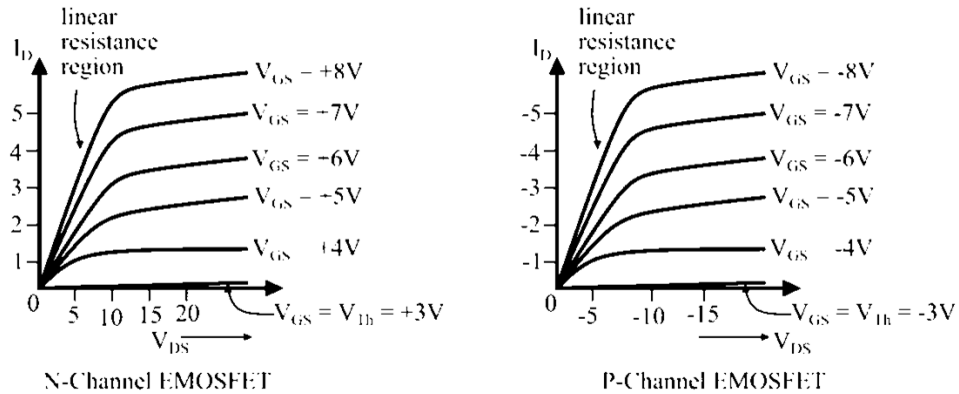


Fig 17: Drain characteristics for N & P type Enhancement MOSFET

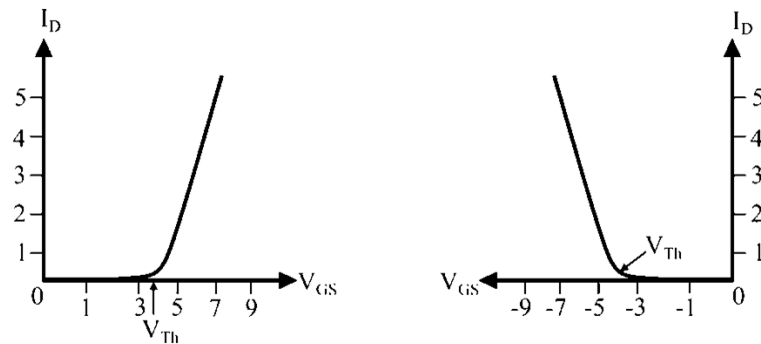


Fig 18: Transfer characteristics for N type Enhancement MOSFET

3.Complementary Metal Oxide Semiconductor Field Effect Transistors (CMOS)

3.1. Construction of CMOS

The term CMOS stands for “Complementary Metal Oxide Semiconductor”. CMOS technology is one of the most popular technologies in the computer chip design industry and broadly used today to form integrated circuits in numerous and varied applications. Today’s computer memories, CPUs and cell phones make use of this technology due to several key advantages. This technology makes use of both P channel and N channel semiconductor devices. One of the most popular MOSFET technologies available today is the Complementary MOS or CMOS technology. This is the dominant semiconductor technology for microprocessors, microcontroller chips, memories like RAM, ROM, EEPROM and application specific integrated circuits (ASICs).

CMOS technology combines two types of transistors, NMOS and PMOS, in a complementary arrangement. This design significantly reduces power consumption because it only draws current when switching between logic states. Unlike older designs that rely on NMOS-only or bipolar junction transistors and consume power continuously, CMOS circuits remain idle (and consume little power) when not switching.

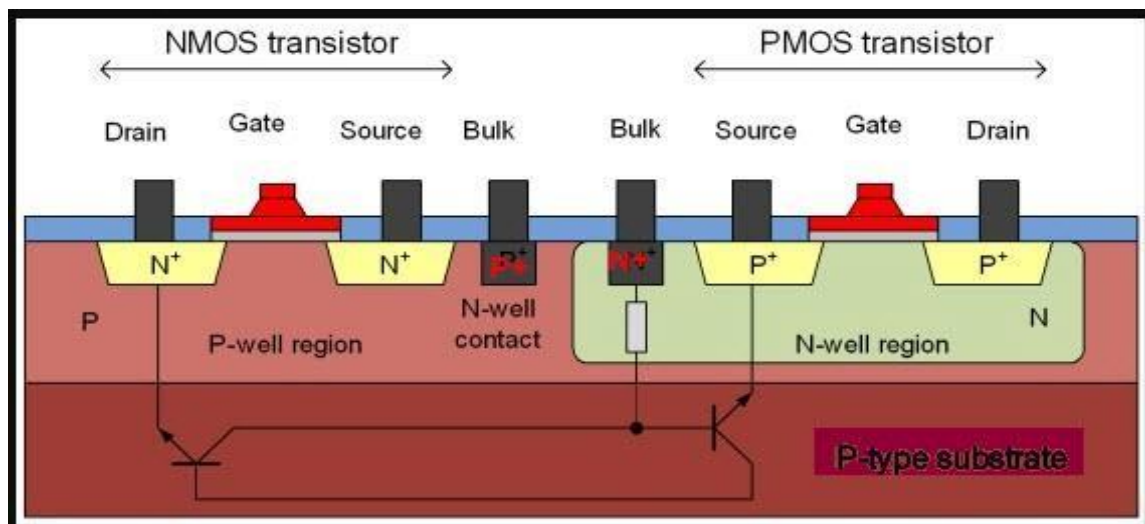
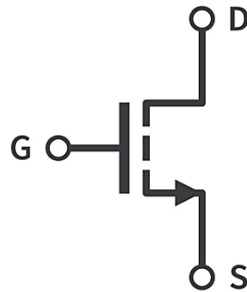


Fig. 2.13.1 CMOS Construction using NMOS & PMOS

NMOS

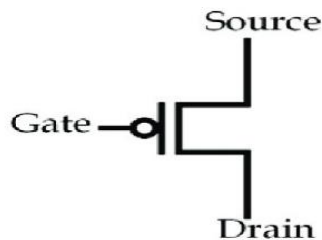
NMOS is built on a p-type substrate with n-type source and drain diffused on it. In NMOS, the majority carriers are electrons. When a high voltage is applied to the gate, the NMOS will conduct.

Similarly, when a low voltage is applied to the gate, NMOS will not conduct. NMOS are considered to be faster than PMOS, since the carriers in NMOS, which are electrons, travel twice as fast as the holes.



PMOS

P- channel MOSFET consists P-type Source and Drain diffused on an N-type substrate. Majority carriers are holes. When a high voltage is applied to the gate, the PMOS will not conduct. When a low voltage is applied to the gate, the PMOS will conduct. The PMOS devices are more immune to noise than NMOS devices.



3.2. CMOS Working Principle & Characteristics

In CMOS technology, both N-type and P-type transistors are used to design logic functions. The same signal which turns ON a transistor of one type is used to turn OFF a transistor of the other type. This characteristic allows the design of logic devices using only simple switches, without the need for a pull-up resistor. In CMOS logic gates a collection of n-type MOSFETs is arranged in a pull-down network between the output and the low voltage power supply rail (V_{ss} or quite often ground). Instead of the load resistor of NMOS logic gates, CMOS logic gates have a collection of p-type MOSFETs in a pull-up network between the output and the higher-voltage rail (often named V_{dd}). Thus, if both a p-type and n-type transistor have their gates connected to the same input, the p- type MOSFET will be ON when the n-type MOSFET is OFF, and vice-versa. The networks are arranged such that one is ON and the other OFF for any input pattern as shown in the figure below.

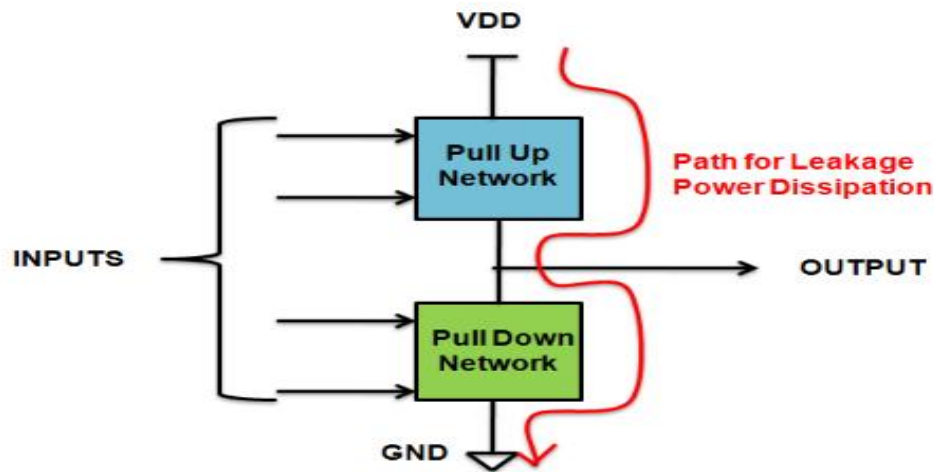


Fig. 2.14.2 CMOS Pull Up & Down Network

3.3. NOT, NAND & NOR implementation using CMOS

CMOS Inverter

A CMOS inverter is a basic building block in digital logic design. It consists of two types of transistors: one PMOS (positive-channel) and one NMOS (negative-channel). These transistors are arranged in a complementary configuration.

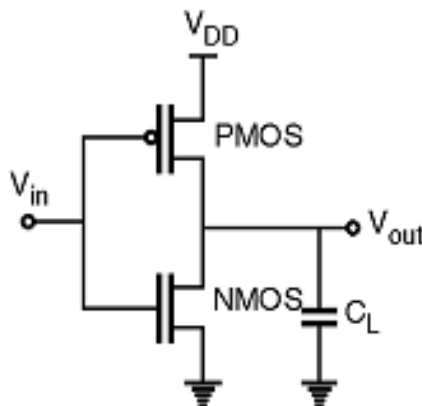


Fig. 2.15.1 CMOS Inverter

Fig. 2.13.2 CMOS Both the PMOS and NMOS transistors are controlled by the same input voltage, called V_{in} , which is connected to the gate terminal of each transistor. The drains of the two transistors are connected. This shared node produces the output voltage, called V_{out} . The source of the PMOS transistor connects directly to the positive power supply, denoted as V_{dd} . The source of the NMOS transistor connects to ground, or 0 volts.

- **When Vin is High (Close to Vdd):** The NMOS transistor conducts (turns on), creating a low-resistance path to ground. The PMOS transistor does not conduct (turns off), effectively disconnecting the output from Vdd. As a result, the output voltage is pulled down to ground. Vout becomes 0 volts, representing logic level 0.

- **When Vin is Low (Close to 0V):** The NMOS transistor turns off and no longer conducts. The PMOS transistor turns on, creating a path from Vdd to the output. The output is pulled up to Vdd. Vout becomes equal to Vdd, representing logic level 1. This switching behaviour causes the CMOS inverter to output the logical opposite (inversion) of the input voltage.

INPUT	LOGIC INPUT	OUTPUT	LOGIC OUTPUT
0 v	0	Vdd	1
Vdd	1	0 v	0

CMOS NAND Gate

The below figure shows a 2-input Complementary MOS NAND gate. It consists of two series NMOS transistors between Y and Ground and two parallel PMOS transistors between Y and VDD. If either input A or B is logic 0, at least one of the NMOS transistors will be OFF, breaking the path from Y to Ground. But at least one of the pMOS transistors will be ON, creating a path from Y to VDD. Hence, the output Y will be high. If both inputs are high, both of the nMOS transistors will be ON and both of the pMOS transistors will be OFF. Hence, the output will be logic low. The truth table of NAND logic gate is given in below table. If either input A or B is logic 0, at least one of the NMOS transistors will be OFF, breaking the path from Y to Ground. But at least one of the pMOS transistors will be ON, creating a path from Y to VDD. Hence, the output Y will be high. If both inputs are high, both of the nMOS transistors will be ON and both of the pMOS transistors will be OFF. Hence, the output will be logic low.

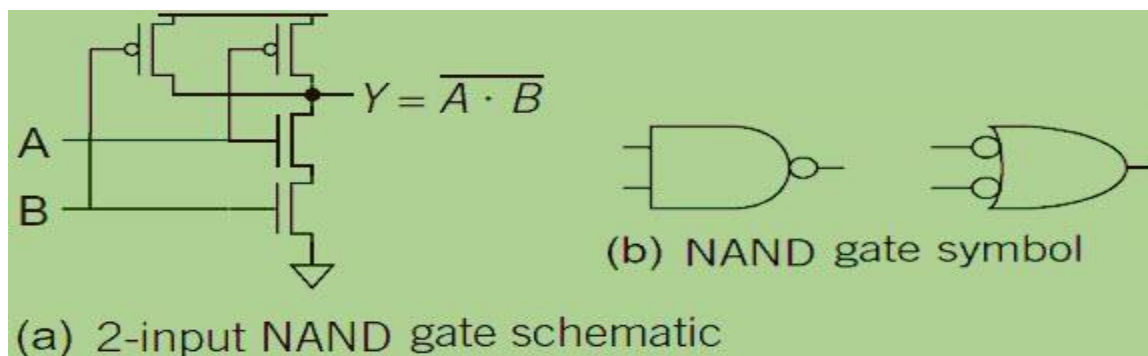


Fig. 2.15.2 CMOS NAND

CMOS NOR Gate

A 2-input NOR gate shown in the figure below. The NMOS transistors are in parallel to pull the output low when either input is high. The PMOS transistors are in series to pull the output high when both inputs are low, as given in below table. The output is never left floating.

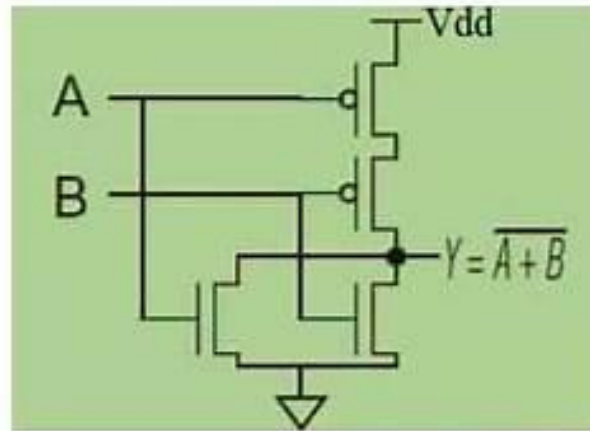


Fig. 2.15.3 CMOS NOR

CMOS Applications

Complementary MOS processes were widely implemented and have fundamentally replaced NMOS and bipolar processes for nearly all digital logic applications. The CMOS technology has been used for the following digital IC designs.

- Computer memories, CPUs
- Microprocessor designs
- Flash memory chip designing
- Used to design application-specific integrated circuits (ASICs)

4. Summary of the Unit

A **Bipolar Junction Transistor (BJT)** is a three-terminal semiconductor device used for amplification and switching. It operates based on the flow of charge carriers (both electrons and holes), making it a bipolar device.

Terminals of BJT

- **Emitter (E):** Injects charge carriers into the base. It is heavily doped.
- **Base (B):** A thin, lightly doped region that controls the flow of carriers.
- **Collector (C):** Collects the charge carriers from the emitter through the base. It is moderately doped.

Types of BJTs

- **NPN Transistor:** Majority carriers are electrons.
- **PNP Transistor:** Majority carriers are holes.

Modes of Operation

- **Cutoff Region:** Transistor is OFF (no current flow). Both junctions are reverse biased.
- **Active Region:** Transistor is ON and amplifies. Base-emitter is forward biased, and base-collector is reverse biased.
- **Saturation Region:** Transistor is fully ON and acts as a closed switch. Both junctions are forward biased.

Configurations of BJT

- **Common Emitter (CE):** High current and voltage gain; used in amplifiers.
- **Common Base (CB):** High voltage gain, low input impedance; used in high-frequency circuits.
- **Common Collector (CC):** High current gain, low output impedance; used for impedance matching.

Current Gain

- γ (Gama): The ratio of emitter current (I_E) to base current (I_B) in the CC configuration
- Typical values: Greater than 100
- β (Beta): The ratio of collector current (I_C) to base current (I_B) in the CE configuration.
- Typical values: 20–200.

α (Alpha): The ratio of collector current (I_C) to emitter current (I_E) in the CB configuration

- Typical values: Close to 1 (0.95–0.99)

Applications

- **Amplifiers:** Audio, RF, and power amplifiers.
 - **Switches:** Used in digital logic circuits.
 - **Oscillators:** For generating periodic waveforms.
 - **Signal Modulation:** Modulation and demodulation applications.
 - **Voltage Regulation:** Stabilizing power supplies.
- A **Field-Effect Transistor (FET)** is a unipolar transistor that controls current flow through a channel using an electric field applied to the gate terminal. It has high input impedance, low power consumption, and is widely used for amplification and switching. A Field-Effect Transistor controls the flow of current using an electric field applied to the gate terminal.
- A **Metal-Oxide-Semiconductor FET (MOSFET)** is a specific type of FET that features an insulated gate, offering even higher input impedance and the ability to operate in both enhancement and depletion modes. MOSFETs are extensively used in digital circuits, power electronics, and high-speed switching due to their efficiency and versatility.

Types of FET:

- JFET (Junction FET): Operates via a p-n junction.
- MOSFET (Metal-Oxide-Semiconductor FET): Uses an insulating oxide layer for gate control.

Control Mechanism:

- Voltage applied to the gate terminal modulates the conductivity of the channel between the source and drain.

Unipolar Device:

- Operates using only one type of charge carrier (electrons in n-channel or holes in p-channel).

High Input Impedance:

- Provides minimal loading on the driving circuit, especially in MOSFETs due to the insulated gate.

Modes of Operation:

- JFET: Always operates in depletion mode (channel is naturally on).
- MOSFET: Operates in both depletion (channel is naturally on) and enhancement modes (channel is naturally off).

Regions of Operation:

- Cut-off Region: No current flows.
- Linear (Ohmic) Region: Acts as a resistor.
- Saturation Region: Acts as a current source.

Applications:

- Amplifiers (low noise, high input impedance).
- Switches in digital circuits.
- Power electronics (e.g., motor drivers, power supplies).

Advantages:

- Low power consumption.
- High speed and efficiency.
- Simple control using voltage.

Disadvantages:

- Sensitive to static discharge (especially MOSFETs).
- Limited current handling in smaller devices.

Key Differences:

Feature	JFET	MOSFET
Gate Structure	p-n junction	Insulated with oxide layer
Modes of Operation	Only depletion	Both depletion and enhancement
Input Impedance	Moderate	Very high
Power Efficiency	Lower due to leakage	Higher due to insulated gate
Applications	Low-noise amplifiers	Switching and high-power circuits

CMOS (Complementary Metal-Oxide-Semiconductor) is a technology used to construct integrated circuits, combining n-type and p-type transistors for efficient logic and memory design. It offers low power consumption, high noise immunity, and is widely used in microprocessors, sensors, and digital cameras. CMOS is the foundation of modern digital electronics.

5. Previously asked Questions in AKTU Exam

5.1. Very Short Answer Questions

Ques1: Explain why BJT is a Bipolar Device & current controlled device? [2023-24]

Ans: BJT is called bipolar device because its operation depends on both majority and minority carriers. A bipolar junction transistor (BJT) is a current-controlled device because the current through two of its terminals is controlled by a current at the third terminal. The base current controls the collector current, and the emitter current is the sum of the base and collector currents.

Ques2: Derive relation between current gains in CB & CE configurations of transistor.

[2022-23 (even sem)]

We know $I_C = \alpha I_E$ — ①

but $I_E = I_B + I_C$, put in Eqⁿ ①

$$\Rightarrow I_C = \alpha (I_B + I_C)$$
$$\Rightarrow I_C = \alpha I_B + \alpha I_C$$
$$\Rightarrow I_C (1 - \alpha) = \alpha I_B$$
$$\Rightarrow \frac{I_C}{I_B} = \frac{\alpha}{1 - \alpha}$$

$\beta = \frac{\alpha}{1 - \alpha}$

 $\longrightarrow$ ②

From Equation 2 :- $\beta (1 - \alpha) = \alpha$

$$\Rightarrow \beta - \alpha \beta = \alpha$$
$$\Rightarrow \alpha (1 + \beta) = \beta$$

$\alpha = \frac{\beta}{1 + \beta}$

Ques3: Explain why FET is a Unipolar Device & Voltage controlled device? [2022-23]

Ans: A field-effect transistor (FET) is a three terminal (namely drain, source and gate) semiconductor device in which current conduction is by only one type of majority carriers (electrons in case of an N-channel FET or holes in a P-channel FET). It is also called the unipolar device. FET is called Voltage controlled device because output is controlled by input voltage. In FET the gate voltage controls the current flow from drain to source.

Ques4: Define Pinch-off voltage for FET. [2022-23]

Ans. The pinch-off condition in a field-effect transistor (FET) occurs when the channel closes and no more current flows between the drain and the source. This happens when the gate voltage is made more negative, which decreases the width of the channel. The voltage at which this occurs is called the pinch-off voltage (V_P). The voltage at which channel will be pinched off i.e. no further movement of carriers from source to drain and the current will be practically constant is called the 'Pinch off voltage'. To operate FET as an amplifier, it is operated in the saturation region.

Ques5: Explain the Transconductance curve of JFET? [2015-16, 2021-22]

OR

Define transconductance of JFET?

Ans: Transconductance (g_m): It is defined the ratio of change in drain current (ΔI_D) and change in gate to source voltage (ΔV_{GS}) at constant drain to source voltage (V_{DS}). Unit of g_m is Siemen.

$$g_m = \left. \frac{\Delta I_D}{\Delta V_{GS}} \right|_{V_{DS} = \text{constant}}$$

$$= \frac{dI_D}{dV_{GS}}$$

Ques 6: What is trans-conductance in FET? what is the relationship between g_m and g_{m0} ? [2017-18]

Ans: Refer above answer

Now

$$g_m = g_{m0} \left(1 - \frac{V_{GS}}{V_{GS(off)}} \right)$$

Ques7: Describe how FET can be used as voltage variable resistor. [2015-2016, 2016-17, 2017-18]

Ans: FET is known as a voltage-variable resistor because its resistance between the drain and source can be controlled by the gate-to-source voltage when operating in the ohmic (linear) region. In this region, the gate voltage modulates the channel conductivity, allowing the resistance to vary inversely with V_{GS} . This makes the FET act as a resistor whose value changes dynamically based on the applied gate voltage, making it suitable for applications like attenuators, mixers, and signal processors. By varying the gate voltage, the channel conductivity changes, which in turn alters the resistance between the drain and source. A higher V_{GS} increases the channel conductivity (decreasing resistance), while a lower V_{GS} reduces conductivity (increasing resistance).

Ques8: List the Differences between JFET and BJT? [2021-22, 2022-23]

	BJT	JFET
1.	It is bipolar device i.e. operation depends on majority and minority carriers both.	It is unipolar device i.e. operation depends on majority carriers only.
2.	Current controlled device i.e. output is controlled by current.	Voltage controlled device i.e. output is controlled by voltage.
3.	Input resistance is very low.	Input resistance is very high.
4.	Temperature dependent due to minority carriers.	Temperature independent due to absence of minority carriers.
5.	Power consumption is high.	Power consumption is low.
6.	Noisier.	Less noisy.
7.	Cost is low.	Cost is high.

Ques9: List the differences between JFET and MOSFET? [2011-12, 2023-24]

	JFET	MOSFET
1.	It can only operate in the depletion mode.	It operates in both depletion mode and enhancement mode.
2.	It has high input impedance; therefore, they are more sensitive towards input voltage signals.	It offers even higher input impedance than the JFETs, therefore they are more resistive.
3.	It is relatively cheaper than MOSFETs	It is too expensive.
4.	These are ideal for low noise applications.	These are mainly used for high noise applications.
5.	Manufacturing process of JFETs is simple.	Manufacturing process of MOSFETs is complex.

5.2. Long Answer Questions

Ques1: Draw and explain the working of PNP Transistor in Common Base Configuration with its characteristic graphs. [2023-24 (odd sem)]
[Refer to 1.6]

Ques2: Discuss the Common Collector configurations of a BJT. [2023-24(even sem)]
[Refer to 1.7.3]

Ques3: Draw the circuit of NPN transistor in CB configuration and discuss its working. Draw input-output characteristics. [2022-23 (odd sem)]
[Refer to 1.6 & 1.7.1]

Ques4: Draw and explain the working of NPN transistor in CE configuration with its suitable characteristics. [2022-23 (even sem)]
[Refer to 1.6 & 1.7.2]

Ques5: Describe the construction and working of an NPN transistor in CE configuration wrt size and doping. Also draw the input and output characteristics graphs. [2021-22 (even sem)]
[Refer to 1.6 & 1.7.2]

Ques6: Define the current gains of CB & CE configurations and derive the relationship between them. [2021-22 (odd sem)]
[Refer to 1.7.4]

Ques7: Draw and explain Common Base NPN transistor with its input and output characteristics graphs. Also write an expression for the output current.
[2021-22 (even sem)]
[Refer to 1.7.1]

Ques8: Give the basic difference between an enhancement and depletion type MOSFET. Discuss the construction of a n channel depletion type MOSFET. Also draw its transfer and drain characteristics.
[2016-2017, 2021-22]
[Refer to 2.8]

Ques9: Explain working principle of Depletion type MOSFET (n-channel). Draw & explain its characteristics.
[2017-2018, 2022-23]
[Refer to 2.8]

Ques10: Explain construction & working principle of N channel JFET (n-channel) with its characteristic's graphs.
[2016-2017, 2017-2018, 2023-24]
[Refer to 2.6]

Ques11: Explain working principle of P channel Depletion MOSFET. Draw with its characteristic's graphs.
[2016-2017, 2023-24]
[Refer to 2.8]

Ques12: Explain working principle of P channel Enhancement MOSFET. Draw with its characteristic's graphs.
[2015-2016]
[Refer to 2.8]

5.3. Numerical Problems

Q1) Determine β if $I_E = 5 \text{ mA}$, $I_C = 4.95 \text{ mA}$

(2021-22/Odd Sem)

Solⁿ: $I_E = I_B + I_C$

$$I_B = I_E - I_C = 5 - 4.95 \Rightarrow \boxed{I_B = 0.05 \text{ mA}}$$

$$\Rightarrow \beta = \frac{I_C}{I_B} = \frac{4.95}{0.05} \Rightarrow \boxed{\beta = 99} \text{ Ans}$$

Q2) A transistor having $\alpha = 0.975$ & reverse saturation current $I_{CBO} = 10 \mu\text{A}$ is operated in CE mode.

If base current is $250 \mu\text{A}$. Calculate I_E & I_C .

(2021-22/Odd Sem)

Solⁿ Given $\alpha = 0.975$, $I_{CBO} = 10 \mu\text{A}$, $I_B = 250 \mu\text{A} = 0.25 \text{ mA}$

$$\beta = \frac{\alpha}{1-\alpha} = \frac{0.975}{1-0.975} \Rightarrow \boxed{\beta = 39}$$

$$\beta = \frac{I_C}{I_B} \Rightarrow I_C = \beta I_B = 39 \times 0.25 \Rightarrow \boxed{I_C = 9.75 \text{ mA}}$$

$$\alpha = \frac{I_C}{I_E} \Rightarrow I_E = \frac{I_C}{\alpha} = \frac{9.75}{0.975} \Rightarrow \boxed{I_E = 10 \text{ mA}} \text{ Ans}$$

Q1) If $\alpha = 0.998$, $I_E = 10\text{mA}$, Calculate I_B & I_C .

Soln we know $\alpha = \frac{I_C}{I_E} \Rightarrow 0.998 = \frac{I_C}{10\text{mA}}$

$$\boxed{I_C = 9.98\text{mA}}$$

$$\Rightarrow I_E = I_B + I_C \Rightarrow I_B = I_E - I_C \\ = 10 - 9.98$$

$$\boxed{I_B = 0.02\text{mA}} \quad \underline{\underline{\text{Ans}}}$$

Q2) For a BJT, $\alpha = 0.992$, $I_C = 10.2\text{mA}$, Calculate β , I_B , I_E .

Soln we know $\beta = \frac{\alpha}{1-\alpha} = \frac{0.992}{1-0.992} \Rightarrow \boxed{\beta = 124}$

$$\Rightarrow \beta = \frac{I_C}{I_B} \Rightarrow I_B = \frac{I_C}{\beta} = \frac{10.2\text{mA}}{124} \Rightarrow \boxed{I_B = 0.0822\text{mA}}$$

or $\boxed{I_B = 82.2\mu\text{A}}$

$$\Rightarrow \boxed{I_E = I_B + I_C} \Rightarrow I_E = 0.0822 + 10.2$$
$$\boxed{I_E = 10.28\text{mA}} \quad \underline{\underline{\text{Ans}}}$$

Q3 In a BJT, $I_C = 18.5 \text{ mA}$, $I_E = 19.8 \text{ mA}$, $I_{CO} = 21 \mu\text{A}$.
Calculate - α , β & I_{CEO} (2)

Soln $\therefore I_E = I_B + I_C$

$$I_B = I_E - I_C = 19.8 \text{ mA} - 18.5 \text{ mA}$$

$$I_B = 1.3 \text{ mA}$$

$$\Rightarrow \beta = \frac{I_C}{I_B} = \frac{18.5 \text{ mA}}{1.3 \text{ mA}} \Rightarrow \beta = 14.2$$

$$\Rightarrow \alpha = \frac{\beta}{1+\beta} \Rightarrow \alpha = \frac{14.2}{1+14.2} \Rightarrow \alpha = 0.93$$

we know $I_{CEO} = (1+\beta) I_{CBO}$
 $= (1+14.2) \times 21 \mu\text{A}$

$$I_{CEO} = 319.2 \mu\text{A} \quad \underline{\text{Ans}}$$

Q4) The value of α changes from 0.980 to 0.989. calculate
 % change in β .

Soln Given $\alpha_1 = 0.980$
 $\alpha_2 = 0.989$

$$\beta_1 = \frac{\alpha_1}{1-\alpha_1} = \frac{0.980}{1-0.980}$$

$$\beta_2 = \frac{\alpha_2}{1-\alpha_2} = \frac{0.989}{1-0.989}$$

$$\% \text{ change in } \beta = \frac{\beta_2 - \beta_1}{\beta_1} \times 100$$

Ans

When a reverse gate voltage of 15 V is applied to a JFET, the gate current is $10^{-3} \mu\text{A}$. Find the resistance between gate and source.

Solution.

$$V_{GS} = 15 \text{ V}; I_G = 10^{-3} \mu\text{A} = 10^{-9} \text{ A}$$

$$\therefore \text{Gate to source resistance} = \frac{V_{GS}}{I_G} = \frac{15 \text{ V}}{10^{-9} \text{ A}} = 15 \times 10^9 \Omega = 15,000 \text{ M}\Omega$$

When V_{GS} of JFET changes from -3.1 V to -3 V , the drain current changes from 1 mA to 1.3 mA . What is the value of transconductance?

Solution.

$$\Delta V_{GS} = 3.1 - 3 = 0.1 \text{ V} \quad \dots \text{magnitude}$$

$$\Delta I_D = 1.3 - 1 = 0.3 \text{ mA}$$

$$\therefore \text{Transconductance, } g_{fs} = \frac{\Delta I_D}{\Delta V_{GS}} = \frac{0.3 \text{ mA}}{0.1 \text{ V}} = 3 \text{ mA/V} = 3000 \mu \text{ mho}$$

A JFET has a drain current of 5 mA . If $I_{DSS} = 10 \text{ mA}$ and $V_{GS}(\text{off}) = -6 \text{ V}$, find the value of
(i) V_{GS} and (ii) V_P .

Solution.

$$I_D = I_{DSS} \left[1 - \frac{V_{GS}}{V_{GS(\text{off})}} \right]^2$$

or

$$5 = 10 \left[1 + \frac{V_{GS}}{6} \right]^2$$

or

$$1 + \frac{V_{GS}}{6} = \sqrt{5/10} = 0.707$$

(i) $\therefore$

$$V_{GS} = -1.76 \text{ V}$$

(ii) and

$$V_P = -V_{GS(\text{off})} = 6 \text{ V}$$

A JFET has the following parameters: $I_{DSS} = 32 \text{ mA}$; $V_{GS}(\text{off}) = -8 \text{ V}$; $V_{GS} = -4.5 \text{ V}$. Find the value of drain current.

Solution :

$$\begin{aligned} I_D &= I_{DSS} \left[1 - \frac{V_{GS}}{V_{GS(\text{off})}} \right]^2 \\ &= 32 \left[1 - \frac{(-4.5)}{-8} \right]^2 \text{ mA} \\ &= 6.12 \text{ mA} \end{aligned}$$

The datasheet of a JFET gives the following information: $I_{DSS} = 3 \text{ mA}$, $V_{GS}(\text{off}) = -6 \text{ V}$ and $g_m(\text{max}) = 5000 \mu\text{S}$. Determine the transconductance for $V_{GS} = -4 \text{ V}$ and find drain current I_D at this point.

Solution. At $V_{GS} = 0$, the value of g_m is maximum i.e. g_{mo} .

$$\therefore g_{mo} = 5000 \mu\text{S}$$

$$\begin{aligned} \text{Now } g_m &= g_{mo} \left(1 - \frac{V_{GS}}{V_{GS}(\text{off})} \right) \\ &= 5000 \mu\text{S} \left(1 - \frac{-4\text{V}}{-6\text{V}} \right) \\ &= 5000 \mu\text{S} \left(\frac{1}{3} \right) = 1667 \mu\text{S} \end{aligned}$$

$$\begin{aligned} \text{Also } I_D &= I_{DSS} \left(1 - \frac{V_{GS}}{V_{GS}(\text{off})} \right)^2 \\ &= 3 \text{ mA} \left(1 - \frac{-4}{-6} \right)^2 = 333 \mu\text{A} \end{aligned}$$

6. Model Questions (Theory & Numerical)

- Draw and explain the working of NPN transistor in Common Base Configuration with suitable characteristics. [Refer to 1.7.1]
- Describe the construction, working and characteristics of N channel JFET. [Refer to 2.6]
- Describe the construction, working and characteristics of P channel DMOS. [Refer to 2.8]
- Describe the construction, working and characteristics of N channel EMOS. [Refer to 2.8]
- Explain how JFET behaves as a VVR and a constant current source. [Refer to 2.5]
- Derive the relation between the current gains of CB & CE configuration. For a transistor $I_E = 10 \text{ mA}$ and $\alpha = 0.987$. Find I_C and I_B . [[Refer to 4.3]
- Draw and explain the working of N-P-N transistor in Common Emitter configuration with its suitable characteristic graph. [Refer to 1.7.2]
- Compare BJT & FET. Determine β_{dc} and I_{CBO} , if $I_E = 6 \text{ mA}$, $I_C = 5.92 \text{ mA}$ and $I_{CEO} = 200 \mu\text{A}$ [Refer to 4.3]
- A transistor having $\alpha = 0.975$ and reverse saturation current $I_{CBO} = 10 \mu\text{A}$ is operated in CE mode. If the base current is $250 \mu\text{A}$. Calculate I_E and I_C . [[Refer to 4.3]
- Compare the following: [Refer to 2.8]
 1. DMOS & EMOS
 2. MOSFET & JFET
- Derive the expression for transconductance for a JFET. [[Refer to 2.7]
- A JFET has $V_{GS(\text{off})} = -20 \text{ V}$ and $I_{DSS} = 12 \text{ mA}$. Plot the transconductance curve for $V_{GS} = 0 \text{ V}, -5 \text{ V}, -10 \text{ V}, -15 \text{ V} \text{ \& } -20 \text{ V}$. [Refer to 4.3]
- For an N Channel JFET: $I_{DSS} = 20 \text{ mA}$, $V_P = -8 \text{ V}$, $g_{mo} = 5000 \mu\text{S}$. Find the value of drain current and transconductance at $V_{GS} = -4 \text{ V}$. [Refer to 4.3]
- Explain the role of p-type MOSFET and n-type MOSFET in CMOS construction.
- Compare the construction of nMOS, pMOS, and CMOS circuits.
- List at least five major applications of CMOS technology.
- Explain how CMOS technology is used to implement logic gates.
- Explain the VTC (Voltage Transfer Characteristic) of a CMOS inverter.